

The decaf order has many nice properties. It is the product of Bruhat order for S_k and the poset determined by pushbacks on the subset $\{w \in \mathcal{W}_{n,k} \mid \pi(w) = id\}$. The decaf order is a ranked poset on $\mathcal{W}_{n,k}$, and its rank generating function is the same as the Poincaré polynomial in (1.4). The medium roast and espresso orders are not ranked posets in general. For $n \geq 5$ and most values of k , there are covering relations in the medium roast Fubini-Bruhat order $(\mathcal{W}_{n,k}, \leq)$ with a dimension difference of 2 or more, causing the medium roast Fubini-Bruhat order to be unranked in general. For example, in $\mathcal{W}_{5,4}$, 44312 covers 41321, but 44312 has dimension 1, and 41321 has dimension 3.

Theorem 4.2. The Superpushback Rule. *Suppose $w \in \mathcal{W}_{n,k}$, $i \in [k-1]$, and $j \in [n]$ such that $w_j = \pi_i$ is a redundant letter in w . If $i + p \leq k$ and v is obtained from w by replacing w_j by $\pi_{i+p}(w)$, then $v \rightarrow w$ and this is a covering relation in both espresso and medium roast orders.*

Theorem 4.3. The Lifting Property. *Suppose $v, w \in \mathcal{W}_{n,k}$, $i \in [k-1]$, $\alpha_{i+1}(v) < \alpha_i(v)$, and $\alpha_{i+1}(w) < \alpha_i(w)$. If $v \leq w$ in medium roast Fubini-Bruhat order, then $s_i v \leq s_i w$. Furthermore, if $v \rightarrow w$, then $s_i v \rightarrow s_i w$.*

5 Essential Sets

We extend the notion of a Rothe diagram from Definition 2.1 to Fubini words. This allows us to define the essential set for a Fubini word. We then show the essential set determines a minimal set of rank equations on the corresponding PR variety, generalizing Fulton's essential set for permutations and Schubert varieties [7]. This leads to an essential set characterization of $v \leq w$ in medium roast order.

Definition 5.1. [16] *A Fubini word $w \in \mathcal{W}_{n,k}$ is called **convex** if $h < j$ and $w_h = w_j$ implies that $w_i = w_j$ for every i such that $h < i < j$. Then the **convexification** of w , denoted by $\text{conv}(w)$, is the unique convex word such that $\pi(\text{conv}(w)) = \pi(w)$ and the content of w and $\text{conv}(w)$ are the same as multisets. The **standardization** of w , denoted $\text{std}(w) \in S_n$, is obtained by replacing the $n - k$ redundant letters of w with $k + 1, k + 2, \dots, n$ from left to right.*

Deduce from Definition 5.1 that two Fubini words $v, w \in \mathcal{W}_{n,k}$ have the same convexification, $\text{conv}(v) = \text{conv}(w)$, if and only if $\pi(v) = \pi(w)$ and they have the same multiset of letters.

Definition 5.2. *Given Fubini word $w \in \mathcal{W}_{n,k}$, define the **diagram** of w to be $D(\text{std}(\text{conv}(w)))$.*

One can observe that $D(\text{std}(\text{conv}(w))) \subset [k] \times [n]$, as none of the bottom $n - k$ rows will contribute any elements to $D(\text{std}(\text{conv}(w)))$. Thus, the diagram of a Fubini word in $\mathcal{W}_{n,k}$ can be drawn as a $k \times n$ grid of dots. For example, the convexification of $w = 44253136541 \in \mathcal{W}_{11,6}$ is 44425533116, and $\text{std}(44425533116) = [4, 7, 8, 2, 5, 9, 3, 10, 1, 11, 6]$. So the diagram for w is $D([4, 7, 8, 2, 5, 9, 3, 10, 1, 11, 6])$. See Figure 1.